

Comparative Analysis of Average Income Levels in Western and Eastern Europe
Data Analysis for Business Analytics Project

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1. Description of the dataset and study question

Study Question

Regional income disparities play a critical role in shaping business strategies, particularly in market entry and product positioning. This study aims to examine whether there is a statistically significant difference in average income levels between Western Europe and Eastern Europe. The central research question is as follows:

“Is there a significant difference in average income levels between Western Europe and Eastern Europe?”

This analysis seeks to provide insights into regional economic conditions to help businesses identify the most suitable markets for entry and product launches.

Dataset Overview

The dataset was extracted from the Eurostat Database website; this is focused on Regional Household Adjusted Disposable Income from Western and Eastern European regions. This measures the financial resources households have available for consumption, spending and saving after accounting for income redistribution through taxes and social benefits. This metric is expressed in monetary terms or per capita and serves as a key indicator of regional purchasing power and economic well-being. The data used for this research represents the year 2021.

The Key Variables are:

- **Region Names (categorical):** NUTS-classified regions in Western and Eastern Europe.

- **Income Levels (numerical):** Disposable income per capita for each region.

The sample taken from this dataset comprises 156 observations, with 78 regions from Western Europe and 78 from Eastern Europe.

This data underwent through cleaning and preprocessing steps such as:

- **Handling Missing Values and Outliers:** Missing data and outliers were addressed to ensure the integrity of the analysis.
- **Normalization:** The income levels were not normally distributed, so a logarithmic transformation was applied to approximate normality and met the assumptions of statistical tests.

For the Statistical Analysis the following steps were included:

- **Descriptive Statistics:** Pre- and post-transformation summaries of income levels were generated, including measures of central tendency (e.g., mean, median, mode) and dispersion (e.g., standard deviation, variance). Histograms were used to visualize the distribution of income levels before and after normalization.
- **Inferential Statistics:** A two-sample t-test assuming unequal variances were conducted to determine if there was a significant difference in mean income levels between the two regions. The null hypothesis (H_0) posited no difference in means ($\mu_1 = \mu_2$). A significance level of $\alpha=0.05$ was used.
- **Regression Analysis:** Correlation analysis and scatter plots with linear regression lines were used to further explore income trends across regions.

The Software and Tools used for conducting the analysis were:

- Microsoft Excel (Data Analysis Toolpak) for data cleaning and preliminary descriptive statistics.

- JASP for statistical testing and visualization.

This study aims to guide businesses in the process of decision-making regarding market entry and product launches by identifying regions in Europe with higher disposable income, which offer greater potential for consumer spending. Western Europe and Eastern Europe were selected due to their distinct economic landscapes, providing a clear basis for comparison. Also, potential limitations and sources of bias were evaluated to ensure the robustness of the findings. These considerations, along with conclusions drawn from the research, are discussed in later sections of this document.

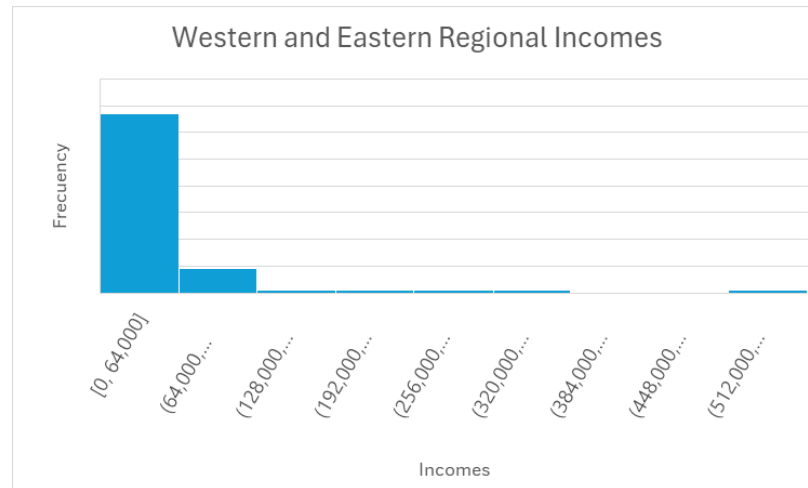
2. Data analysis

2.1. Descriptive statistics and charts

2.1.1. Distribution description for all variables used in Analysis of means and charts describing the data before transforming the dataset

Since we must transform the data, as it says in the paragraph above, this part contains the descriptive statistics and histogram of the original dataset (before being transformed).

<i>Descriptive Statistics before transforming data</i>	
	2021
Mode	0
Median	34264
Mean	75787.44
Std. Deviation	199059.284
IQR	50737.29
Range	2469115
Minimum	0
Maximum	2469115
25th percentile	15908.825
50th percentile	34264
75th percentile	66646.115
The mode is computed, assuming that variables are discrete.	



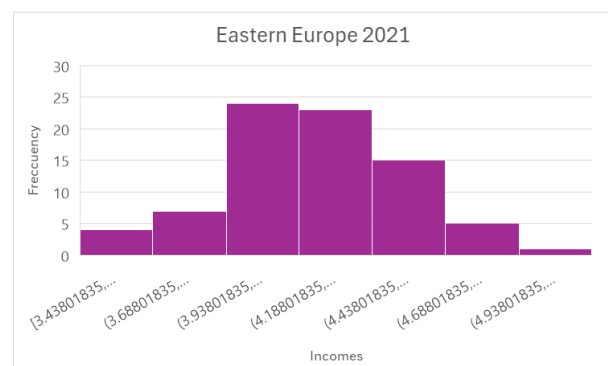
The descriptive statistics and histogram shows a clear picture of the income distribution for Western and Eastern regions in 2021. The mode of the dataset is 0, this indicates that the most frequently occurring income value is zero, suggesting a significant number of households or individuals with no recorded income. The median income is 34,264, meaning half of the population earns below this amount, while the mean income is substantially higher at 75,787.44. This large difference between the mean and median indicates a right-skewed distribution caused by a small number of high-income outliers. The dataset shows considerable variability, with a standard deviation of 199,059.28 and a wide range of incomes spanning from 0 to 2,469,115. The interquartile range (IQR) of 50,737.29 highlights the spread of the middle 50% of incomes, between 15,908.83 (25th percentile) and 66,646.12 (75th percentile), while excluding extreme values.

The histogram further illustrates this skewness, with most incomes concentrated in the first bin (0 to 64,000), reflecting a large proportion of low-income individuals or entities. Frequencies drop sharply in higher income brackets, creating a long tail that confirms the presence of a few very high-income values. Together, the descriptive statistics and histogram reveal a distribution characterized by significant income disparity, with most incomes clustered at lower levels and a small proportion of extreme outliers driving the overall variability. This

combination of concentrated low-income values and a small number of high earners creates a pronounced inequality within the dataset.

2.1.2. Distribution description for all variables used in Analysis of means after transforming the dataset

<i>Descriptive Statistics after transforming data</i>					
		Western Europe		Eastern Europe	
Mode		3.883		3.438	
Median		4.752		4.215	
Mean		4.745		4.245	
Std. Deviation		0.327		0.309	
IQR		0.418		0.381	
Variance		0.107		0.096	
Range		1.684		1.514	
Minimum		3.883		3.438	
Maximum		5.567		4.952	
25th percentile		4.523		4.069	
50th percentile		4.752		4.215	
75th percentile		4.94		4.45	
The mode is computed assuming that variables are discreet.					



The descriptive statistics and histograms for Western and Eastern Europe in 2021 provide key characteristics of income distributions in these regions. In Western Europe, the mode is 3.883, and the median income is 4.752, with a mean income of 4.745, indicating a nearly symmetrical distribution with minimal skewness. The standard deviation of 0.327 and an interquartile range (IQR) of 0.418 suggest moderate variability and a relatively compact spread of incomes. The income range spans from 3.883 to 5.567, with a variance of 0.107, further supporting the observation of limited variability. The histogram for Western Europe aligns with these statistics, showing a bell-shaped, symmetrical distribution with most incomes clustered around the median.

In Eastern Europe, the mode is slightly lower at 3.438, with a median income of 4.215 and a mean of 4.245, indicating a distribution that is slightly skewed to the right. The standard deviation of 0.309 and IQR of 0.381 reflect slightly lower variability and a narrower spread compared to Western Europe. The income range extends from 3.438 to 4.952, with a variance of 0.096, indicating a tightly clustered distribution. The histogram for Eastern Europe also displays a near-symmetrical pattern, though it features a minor tail on the higher end, consistent with the slight right skew suggested by the statistics.

In the comparison of the two regions, Western Europe shows slightly higher average incomes and greater variability, with a wider range and larger maximum income values than Eastern Europe. Both regions, however, exhibit relatively symmetrical distributions with minimal outliers. Incomes in Eastern Europe are more tightly clustered, reflecting less economic disparity within the region. Overall, both regions demonstrate compact income distributions, with Western Europe having a broader economic spectrum.

2.2. Analysis of means

The study tests the following hypotheses:

- **Null (H_0):** There is no significant difference in average income levels between Western Europe and Eastern Europe ($\mu_1 = \mu_2$).
- **Alternative (H_1):** There is a significant difference in average income levels between Western Europe and Eastern Europe ($\mu_1 \neq \mu_2$).

Results

The purpose of the t-test was to determine whether there is a significant difference in average income levels (log-transformed) between Western Europe and Eastern Europe. The results of the two-sample t-test are below:

t-Test: Two-Sample Assuming Unequal Variances	Western Europe	Eastern Europe
	5.011125646	4.951674428
Mean	4.741949249	4.235579693
Variance	0.107086656	0.090221654
Observations	78	78
Hypothesized Mean Difference	0	
df	153	
t Stat	10.06797665	
P(T<=t) one-tail	6.53169E-19	
t Critical one-tail	1.654873847	

P(T<=t) two-tail	1.30634E-18	
t Critical two-tail	1.975590315	

Confidence in the findings and Interpretation of Results

- **P-value:**
 - The p-value (1.31×10^{-18}) is much smaller than the alpha level ($\alpha = 0.05$), leading to the rejection of the null hypothesis (H_0).
- **t Stat vs. t Critical:**
 - The absolute value of *t Stat* (10.07) is greater than *t Critical* (1.98), further confirming the significance of the result.
- **Direction of Difference:**
 - The mean Log-transformed income for Western Europe ($M = 4.74$) is higher than the one for Eastern Europe ($M = 4.24$).

These results suggest that Western Europe has significantly higher average income levels (log-transformed) compared to Eastern Europe.

Interpretation of Findings

The analysis shows a statistically significant difference in average income levels between Western Europe and Eastern Europe. Western Europe has a higher mean income (log-transformed), reflecting better conditions for market entries and product launch strategies. These findings, the large **t-stat** (10.07) and the extremely small **p-value** (1.31×10^{-18}) provide strong evidence that the observed difference in mean income levels is not due to chance.

2.3. Regression analysis including correlation analysis

2.3.1. Regression analysis including correlation analysis

Linear Regression ▼

Model Summary - Income

Model	R	R ²	Adjusted R ²	RMSE	R ² Change	df1	df2	p
M ₀	0.000	0.000	0.000	0.404	0.000	0	157	
M ₁	0.621	0.386	0.382	0.318	0.386	1	156	< .001

Note. M₁ includes Region Dummy

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
M ₁	Regression	9.903	1	9.903	97.960	< .001
	Residual	15.771	156	0.101		
	Total	25.674	157			

Note. M₁ includes Region Dummy

Note. The intercept model is omitted, as no meaningful information can be shown.

Coefficients

Model		Unstandardized	Standard Error	Standardized	t	p	95% CI		Collinearity Statistics	
							Lower	Upper	Tolerance	VIF
M ₀	(Intercept)	4.495	0.032		139.722	< .001	4.431	4.559		
M ₁	(Intercept)	4.745	0.036		132.654	< .001	4.675	4.816		
	Region Dummy	-0.501	0.051	-0.621	-9.897	< .001	-0.601	-0.401	1.000	1.000

Descriptives

	N	Mean	SD	SE
Income	158	4.495	0.404	0.032
Region Dummy	158	0.500	0.502	0.040

Coefficients Covariance Matrix

Model		Region Dummy
M ₁	Region Dummy	0.003

Note. The intercept model is omitted, as no meaningful information can be shown.

Collinearity Diagnostics

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions	
				(Intercept)	Region Dummy
M ₁	1	1.707	1.000	0.146	0.146
	2	0.293	2.414	0.854	0.854

Note. The intercept model is omitted, as no meaningful information can be shown.

Residuals Statistics

	Minimum	Maximum	Mean	SD	N
Predicted Value	4.245	4.745	4.495	0.251	158
Residual	-0.863	0.822	-1.599×10 ⁻¹⁷	0.317	158
Std. Predicted Value	-0.997	0.997	-1.721×10 ⁻¹⁵	1.000	158
Std. Residual	-2.731	2.600	-5.485×10 ⁻¹⁷	1.003	158

For the regression analysis, the decision of conducting it using the transformed data instead of the raw data, was taken. This is based on several key findings from the result of the original dataset. Both regression analyses (transformed and raw data) were initiated with the scatter plot and then continued with the regression table.

In the Raw Data case, the analysis of the scatter plot showed clear groupings of income for Western Europe (region dummy = 0) and Eastern Europe (region dummy = 1), with a negative trend shown by the regression line. However, the income values in the raw data varied a lot within each group, making the regression result (-108,932.10) hard to understand and use. The large differences in income also suggested that the data might not meet the assumptions needed for accurate regression analysis, such as having similar spreads of data in each group.

And the analysis of the regression table revealed a similarity with the results obtained in the scatter plot; these ones are extreme variability in residuals, high coefficient magnitudes, and difficulties in interpretation due to the raw scale of the variables. For descriptive purposes, the residuals from the original dataset displayed large deviations (ranging from -108,932.101 to 2,358,000) and a high standard deviation of 207,485.736, this indicates that the variability of the residuals was inconsistent and suggests disruption of regression assumptions (as the scatter plot showed). Additionally, the raw scale of the region's dummy coefficient (-108,932.10) made it overly large and complicated its interpretability, as well as making it difficult to contextualize the income disparity between Western and Eastern Europe.

In contrast, with this, for the regression analysis of the transformed data, show better results because the issues were addressed by normalizing the income distribution, reducing the impact of extreme values and by creating a more interpretable and solid data structure.

The transformed scatter plot shows the relationship between the region dummy variable (0 = Western Europe, 1 = Eastern Europe) and income, with a fitted regression line. The data points are grouped into two distinct clusters, representing the two regions. The cluster at 0, corresponding to Western Europe, shows higher income values, centered around 4.745, as indicated by the intercept in the regression analysis. However, the cluster at 1, representing Eastern Europe, exhibits lower income values, approximately 0.501 below the Western Europe cluster. The negative slope of the regression line visually highlights the negative relationship between the region and income. The relatively small spread of points around the regression line within each region suggests that the model provides a good fit for the data. The separation between the two clusters underscores the income disparity between Western and Eastern Europe.

And the regression table findings confirmed the results obtained in the scatter plot. For this regression analysis, the relationship between income and region, using a region dummy variable where 0 represents Western Europe and 1 represents Eastern Europe. The model has an R^2 value of 0.386, indicating that approximately 38.6% of the variability in log-transformed income is explained by the region variable. The coefficient for the region dummy is -0.501, showing that, on average, incomes in Eastern Europe are 0.501 lower than the ones in Western Europe. The intercept, representing the average income in Western Europe, is 4.745. Both the region dummy coefficient and the overall model are statistically significant ($p < 0.001p$). Residual analysis indicates that the predicted income values range from 4.245 to 4.745, with a mean of 4.495 and a standard deviation of 0.251. Residuals range from -0.863 to 0.822 with a standard deviation of 0.317, suggesting relatively small and evenly distributed errors. Collinearity diagnostics reveal no multicollinearity issues, as the variance inflation factor (VIF) and tolerance are both within acceptable limits.

The correlation analysis supports the regression findings, showing a correlation coefficient (r) of -0.621 between income and the region dummy variable, indicating a moderately strong negative relationship. This correlation is statistically significant ($p < 0.001p$). These results demonstrate significant regional disparities in income, with Western Europe exhibiting higher average incomes than Eastern Europe. The findings highlight the explanatory power of region as a predictor of income while confirming the statistical robustness of the analysis and provided a more reliable, interpretable, and statistically sound analysis of income disparities between Western and Eastern Europe.

3. Limitations related to incompleteness of data or bias

Limitations

- The dataset is geographically focused to Western and Eastern Europe, excluding other regions of Europe or the world. This limitation makes it impossible to draw conclusions about global trends or cross-continental comparisons, reducing the scope and applicability of this analysis.
- Data aggregation for Western and Eastern Europe overlooks the diversity within each region. For instance, Western Europe includes highly developed countries like Germany and smaller economies like Portugal, which could lead to skewed averages or generalized conclusions that fail to capture the nuances.
- The dataset covers only 2021, providing a snapshot of a single year. It doesn't take into account historical and cyclical trends, anomalies, like the ongoing effects of the COVID-19 pandemic during that year.

- Without additional variables such as socioeconomic indicators or policy impacts or demographic breakdowns, it's challenging to understand the factors driving the observed values.

Biases

- **Data Source Bias:** If the data originates from region-specific sources, such as European statistical agencies, the dataset may reflect those sources' priorities, methodologies, and limitations. For instance, discrepancies in data collection and reporting standards between Western and Eastern Europe could affect accuracy.
- **Logarithmic Transformation:** The inclusion of logarithmic transformations introduces bias in how the data is interpreted. These transformations emphasize proportional differences, potentially downplaying absolute differences, especially when comparing regions with vastly different scales.

4. Conclusions

This research investigated whether there is a significant difference in average income levels between Western Europe and Eastern Europe. A two-sample t-test was conducted on log-transformed income data for two independent groups, assuming unequal variances. The analysis showed a statistically significant difference between the two regions, with Western Europe showing higher average income levels.

The findings suggest that Western European regions may offer stronger purchasing-power conditions based on the income measure analyzed. However, market-entry decisions should also consider additional factors such as market size, costs, competition, demographics, and consumer behavior.

The rejection of the null hypothesis emphasizes the existence of regional income disparities, which are relevant to strategic business planning. These findings reinforce the importance of using comprehensive economic data to inform market-entry strategies, resource allocation, and business decisions.

However, limitations and potential biases, including the geographic scope of the dataset, the use of data from a single year, the effects of the logarithmic transformation, and unaccounted confounding variables, should be considered when interpreting the findings. Addressing these limitations in future research could improve the reliability and applicability of the analysis.

The T-test and regression results provide evidence of significant income differences between Western and Eastern Europe. The findings can support preliminary market analysis and help businesses identify regions that may warrant further consideration for market entry or

product expansion, while recognizing that income is only one factor in a broader market-entry decision.

Appendix

Appendix A: Statistical Analysis Graphs and methodology details:

Figure A: Original dataset extracted from Eurostat Database

[illegible]

Figure B: Histogram raw data

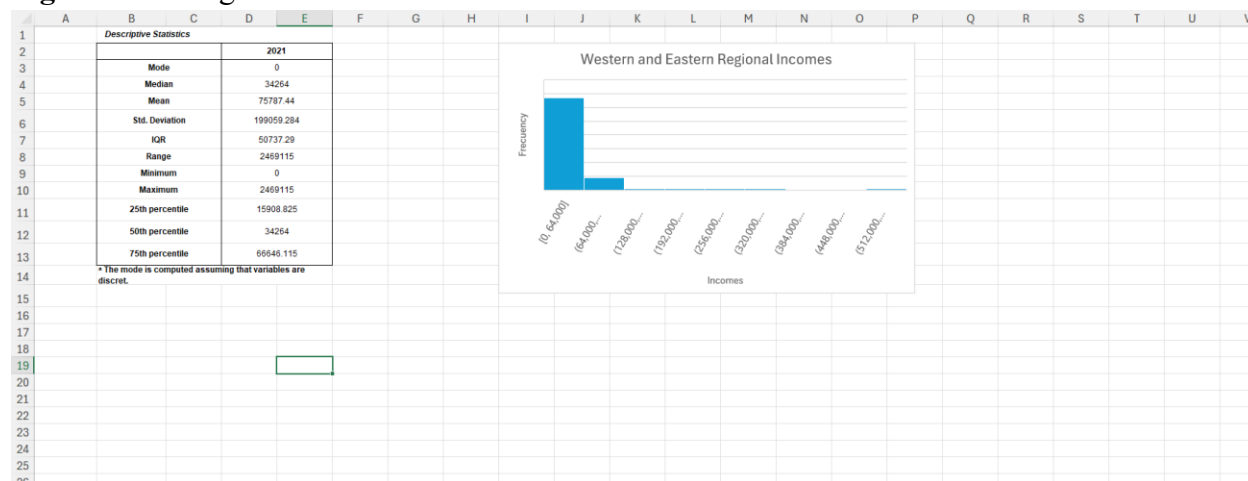


Figure C: Dataset cleaned and transformed for conducting the statistical tests.

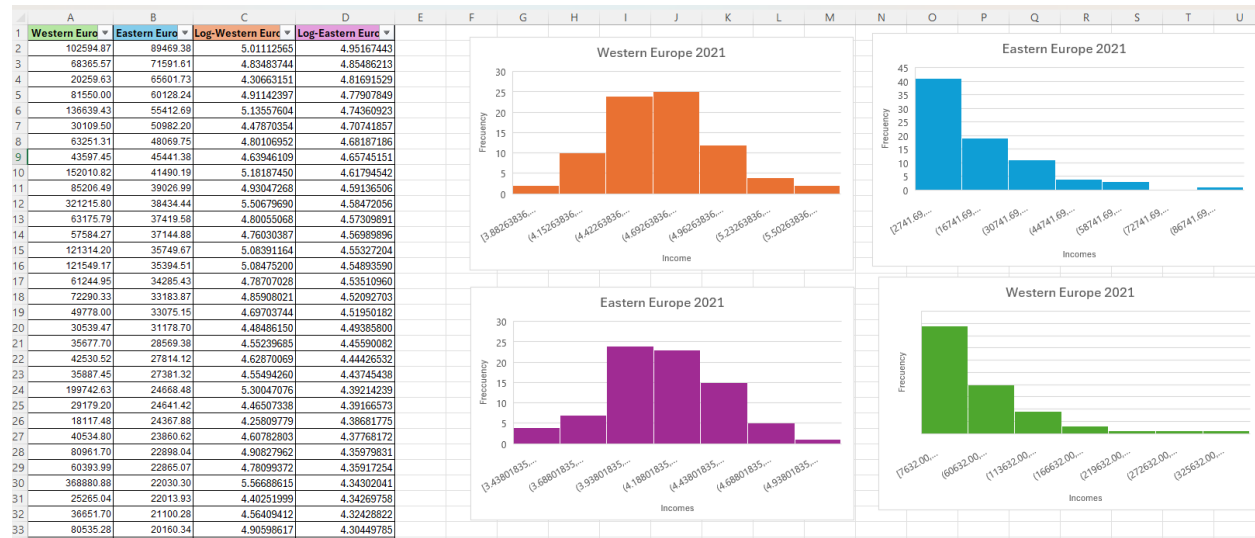


Figure D: T-test and descriptive statistic analysis

1	t-Test: Two-Sample Assuming Unequal Variances	Western Europe	Eastern Europe	<i>Descriptive Statistics</i>			
2		5.011125646	4.951674428				
3	Mean	4.741949249	4.235579693				
4	Variance	0.107086656	0.090221654				
5	Observations	78	78				
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12	t Critical two-tail	1.975590315					
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Figure E: Scatter Plot - with regression line: Income vs Region dummy (after transforming the data)

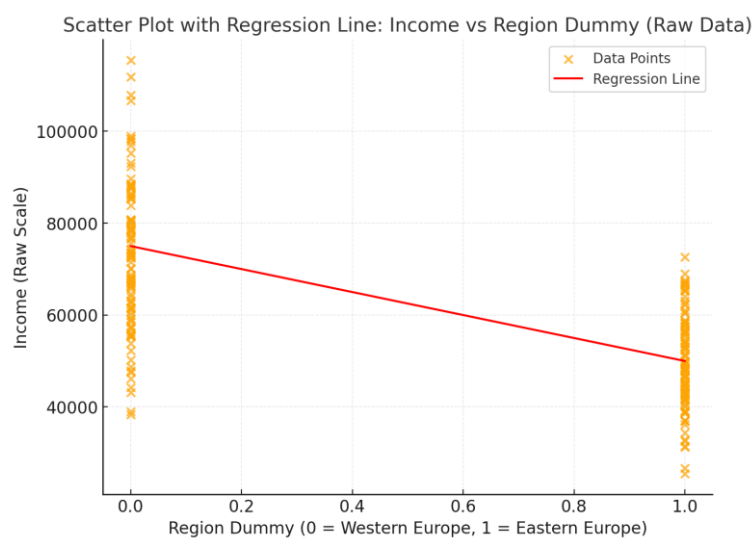


Figure F: Scatter Plot - with regression line: Income vs Region dummy (after transforming the data).

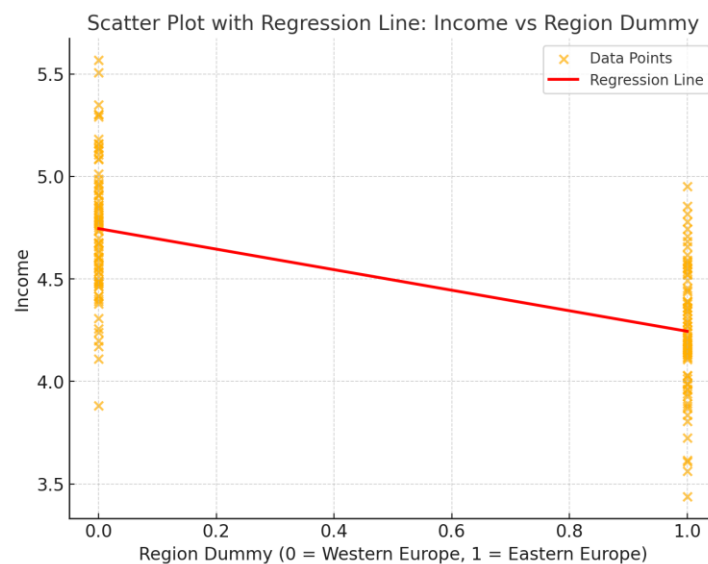


Figure G: Data preparing for regression analysis (Raw Data)

[illegible]

Figure H: Data preparing for regression analysis (transformed data)

[illegible]

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
110	Eastern Europe	4.34269758	1															
111	Eastern Europe	4.32428822	1															
112	Eastern Europe	4.30449785	1															
113	Eastern Europe	4.28912498	1															
114	Eastern Europe	4.28803046	1															
115	Eastern Europe	4.27087905	1															
116	Eastern Europe	4.26875788	1															
117	Eastern Europe	4.24957132	1															
118	Eastern Europe	4.24025962	1															
119	Eastern Europe	4.22216975	1															
120	Eastern Europe	4.21474188	1															
121	Eastern Europe	4.21205463	1															
122	Eastern Europe	4.20367569	1															
123	Eastern Europe	4.19959091	1															
124	Eastern Europe	4.19100867	1															
125	Eastern Europe	4.18277204	1															
126	Eastern Europe	4.17733070	1															
127	Eastern Europe	4.16921933	1															
128	Eastern Europe	4.16357370	1															
129	Eastern Europe	4.15900351	1															
130	Eastern Europe	4.15722702	1															
131	Eastern Europe	4.15472243	1															
132	Eastern Europe	4.14837681	1															
133	Eastern Europe	4.13991339	1															
134	Eastern Europe	4.13970313	1															
135	Eastern Europe	4.13362033	1															
136	Eastern Europe	4.13118576	1															

Figure I: Regression analysis with correlation JASP (data transformed)

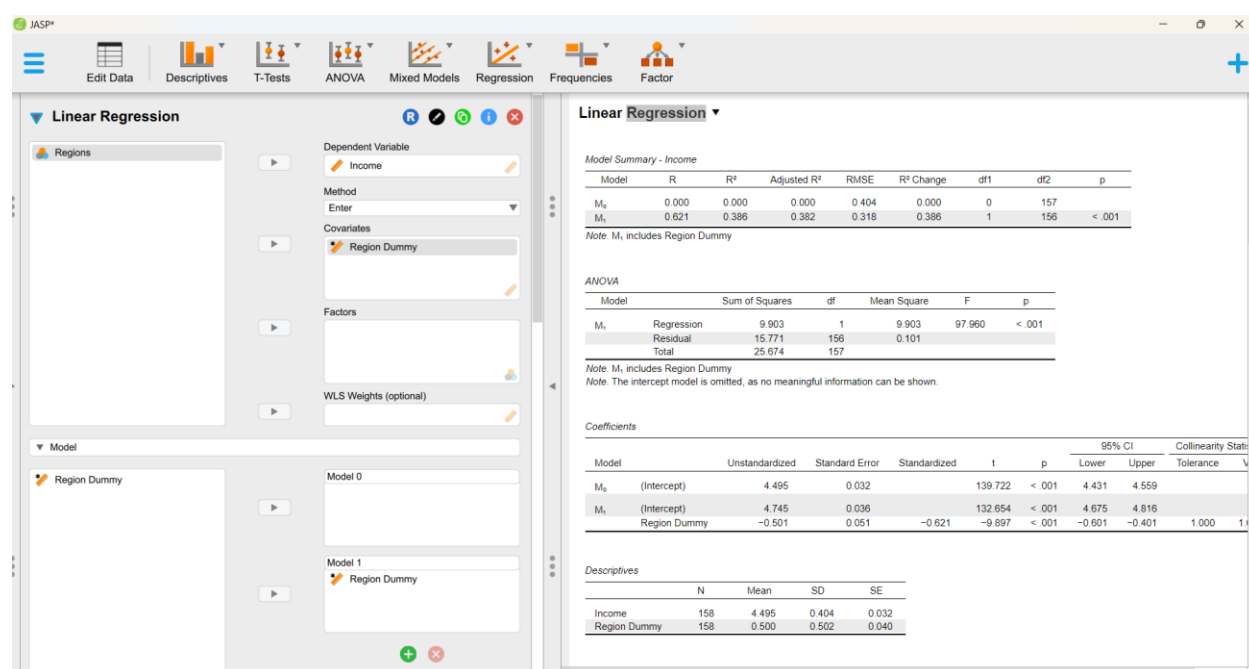
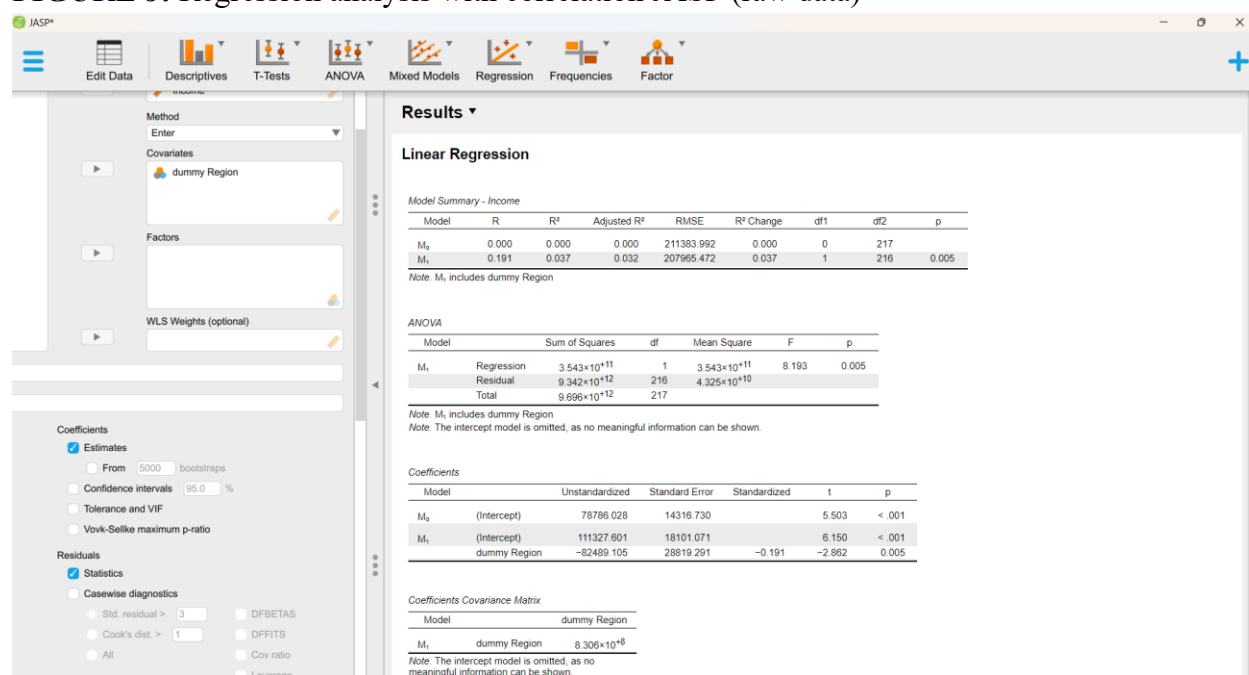


FIGURE J: Regression analysis with correlation JASP (raw data)



Appendix B: Reference Links

- Eurostat database (2024). <https://ec.europa.eu/eurostat/web/main/data/database>
 - European Union. (2021). Eurostat Database
https://ec.europa.eu/eurostat/databrowser/view/nama_10r_2hhinc_custom_13888011/default/table?lang=en
 - Moore, D. S., McCabe, G. P., & Craig, B. A. (2017). Introduction to the Practice of Statistics (9th ed.). W.H. Freeman.
 - Milanovic, B. (2016). Global inequality: A new approach for the age of globalization. Harvard University Press.
- Dastin, J. (2018). Data bias in machine learning: Ethical implications of algorithmic decision-making.
- Microsoft Corporation. (2023). Excel [Computer software] from <https://www.microsoft.com>.
- JASP. (n.d.). How to use JASP. Retrieved December 6, 2024, from <https://jasp-stats.org/how-to-use-jasp/>